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Method for controlling a wind turbine during yaw operation

Abstract

The present disclosure relates to methods (**100, 200, 300**) for controlling a wind turbine (**10**) during a yaw operation. The present disclosure further relates to control systems (**92**) for wind turbines and to wind turbines (**10**). A method for operating a wind turbine (**10**) during yaw operation comprises operating one or more yaw drives (**35**) to rotate the nacelle (**16**) with respect to the tower (**15**). In addition, the method comprises predicting an end of the yaw operation and, in response to predicting an end of the yaw operation, actuating a hydraulic brake (**94**) before the predicted end.

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Background/Summary

(1) The present disclosure relates to wind turbines, and more particularly, to yaw systems and methods for controlling yaw systems.

BACKGROUND

(2) Modern wind turbines are commonly used to supply electricity into the electrical grid. Wind turbines of this kind generally comprise a tower and a rotor arranged on the tower. The rotor, which typically comprises a hub and a plurality of blades, is set into rotation under the influence of the wind on the blades. Said rotation generates a moment that is normally transmitted through a rotor shaft to a generator, either directly (“directly driven” or “gearless”) or through the use of a gearbox. This way, the generator produces electricity which can be supplied to the electrical grid.

(3) The wind turbine hub may be rotatably coupled to a front of the nacelle. The wind turbine hub may be connected to a rotor shaft, and the rotor shaft may then be rotatably mounted in the nacelle using one or more rotor shaft bearings arranged in a frame inside the nacelle. The nacelle is a housing arranged on top of a wind turbine tower that may contain and protect the gearbox (if

present) and the generator (if not placed outside the nacelle) and, depending on the wind turbine, further components such as a power converter, and auxiliary systems.

(4) Wind turbines usually comprise a yaw system for rotating the rotor and the nacelle with respect to the tower about a longitudinal axis of the tower.

(5) During power production operation, the direction of the wind may change. When the wind turbine rotor is misaligned with respect to the wind direction, the yaw system rotates the nacelle about a longitudinal axis of the tower to reach an alignment with the wind. The determination that the nacelle and rotor are no longer aligned with the prevailing wind direction may be made based e.g. on a weather vane mounted on the nacelle. However, other methods based on measuring loads and/or oscillations are also known. Generally, the yaw system may be activated if the prevailing wind direction deviates from the rotor and nacelle orientation by more than a predefined threshold (e.g. 5°, 7°, or 10° or more) for at least a predetermined period of time. A predetermined period of time may be 1 minute, 5 minutes, 10 minutes or more. Seconds to minute averages, e.g. 3 second, 10 second, 1 minute, or 5 minute averages, may for example be determined to calculate the direction of the prevailing wind speed.

(6) Apart from power production, other modes of operation may also involve rotation of the nacelle about the longitudinal axis of the tower. A non-limiting example is a cable untwist operation. Wind turbines comprise cables (power cables and/or other cables) that may be fixed at their ends, e.g. from a component in the nacelle to some other component at some point in the tower. As a result of the previously mentioned rotation during power production operation, the cables may become twisted. Generally, a twist limit exists beyond which the cables may be damaged. Before such limit is reached, the cables need to be untwisted. To this end, an untwist operation of a wind turbine may comprise stopping the wind turbine and yawing the nacelle until a certain predetermined orientation is reached in which the cables are not substantially twisted.

(7) The yaw system normally performs this rotation of the nacelle by means of a yaw drive that includes a plurality of motors, e.g. electric or hydraulic motors, with suitable gearboxes for driving gears (pinions) that mesh with an annular gear or gear ring attached to the nacelle or to the wind turbine tower. The nacelle can thus be rotated around the tower's longitudinal axis in or out of the wind direction. The rotatable connection between the wind turbine tower and the nacelle is called a yaw bearing. The yaw bearing can be of the roller or gliding type.

(8) Normally, when the rotor is positioned according to the desired orientation, e.g. aligned with the wind direction in case of power production operation, the yaw system maintains the orientation by means of brakes, e.g. hydraulic brake calipers and/or motor brakes of the yaw motors. These brakes can be activated and deactivated by the wind turbine control system.

(9) In particular, it is known for the wind turbine controller to activate and deactivate the different components in a defined sequence during a yaw operation when the desired orientation is reached. Thus, a stop command is provided to the motors once a desired orientation is reached and a command is sent to activate a yaw brake system, e.g. a hydraulic brake. Furthermore, it is known to send a command to the motor brakes of the yaw motors after some time to also block the motors.

(10) It has surprisingly been found that in some situations, particularly in case of high winds and/or wind gusts, the aforementioned operation can lead to misalignment, and premature wear and tear of the yaw system.

(11) The present disclosure seeks to provide improved methods of operating a wind turbine and wind turbines that at least partially reduce one or more of the aforementioned problems.

SUMMARY

(12) In an aspect of the present disclosure, a method for operating a wind turbine during a yaw operation is provided. The method includes operating one or more yaw drives to rotate the nacelle with respect to the tower. The method also includes predicting an end of the yaw operation and, in response to predicting the end of the yaw operation, actuating a hydraulic brake before the predicted end.

(13) According to this aspect, the yaw operation sequence is such that actuation of the hydraulic brake is carried out in response to predicting an end of a yaw operation and not at or after said end. That is, the operating state of the hydraulic brake is changed in response to predicting an end of the yaw operations. The actuation is done in anticipation of the predicted end so that a suitable braking moment, is applied by the time the end of the yaw operation is reached. The braking moment mitigates potential slipping of the nacelle when yawing operation is stopped, which can result in uncontrolled movement of the nacelle. With this operation, the risk of damage, failure or premature wear of wind turbine components is mitigated.

(14) In a further aspect of the disclosure, another example of a method for operating a wind turbine during a yaw operation is provided. The method includes operating one or more yaw drives to rotate the nacelle with respect to the tower. The method includes predicting an end of the yaw operation and actuating a hydraulic brake before the predicted end so that the braking moment exerted by the hydraulic brake at the predicted end of the yaw operation is at least 60%, specifically at least 70% and more specifically at least 80% of its rated value.

(15) In a further aspect of the disclosure, a control system for a wind turbine is provided. The control system is configured to perform methods according to any example of the aforementioned aspects. The control system is configured to instruct one or more yaw drives to rotate the nacelle of a wind turbine with respect to the tower during a yaw operation and to predict an end of the yaw operation. The control system is further configured to control, in response to predicting an end of the yaw operation, the actuation of a hydraulic brake before the predicted end.

(16) Still in a further aspect of the disclosure, a wind turbine is provided. The wind turbine comprises a tower, a nacelle rotatably mounted on the tower, and a yaw system for rotating the nacelle with respect to the tower around a longitudinal axis of the tower. The wind turbine furthermore comprises a control system according to previous example of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a perspective view of one example of a wind turbine;
- (2) FIG. 2 illustrates a simplified, internal view of one example of the nacelle of the wind turbine of the FIG. 1;
- (3) FIG. 3 schematically illustrates one example of a yaw system;
- (4) FIG. 4 shows a flow chart of an example of a method for controlling a wind turbine during a yaw operation;
- (5) FIG. 5 schematically illustrates the operation of the yaw system during a yaw operation according to an example;
- (6) FIG. 6 shows a flow chart of another example of a method for controlling a yaw motor brake during a yaw operation of the wind turbine;
- (7) FIG. 7 schematically illustrates the operation of the yaw system during a yaw operation according to another example;
- (8) FIG. 8 shows a flow chart of a further example of a method for controlling a yaw motor brake during a yaw operation of the wind turbine;
- (9) FIG. 9 schematically illustrates the operation of the yaw system during a yaw operation according to a further example; and
- (10) FIG. 10 schematically illustrates the operation of the yaw system during a yaw operation according to the prior art.

DETAILED DESCRIPTION OF EXAMPLES

- (11) Reference now will be made in detail to embodiments, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation only, not as a

limitation. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure. For instance, features illustrated or described as part of one example can be used with another example to yield a still further example. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(12) FIG. 1 is a perspective view of an example of a wind turbine **10**. In the example, the wind turbine **10** is a horizontal-axis wind turbine. Alternatively, the wind turbine **10** may be a vertical-axis wind turbine. In the example, the wind turbine **10** includes a tower **15** that extends from a support system **14** on a ground **12**, a nacelle **16** mounted on tower **15**, and a rotor **18** that is coupled to nacelle **16**. The rotor **18** includes a rotatable hub **20** and at least one rotor blade **22** coupled to and extending outward from the hub **20**. In the example, the rotor **18** has three rotor blades **22**. In an alternative embodiment, the rotor **18** includes more or less than three rotor blades **22**. The tower **15** may be fabricated from tubular steel to define a cavity (not shown in FIG. 1) between a support system **14** and the nacelle **16**. In an alternative embodiment, the tower **15** is any suitable type of a tower having any suitable height. According to an alternative, the tower can be a hybrid tower comprising a portion made of concrete and a tubular steel portion. Also, the tower can be a partial or full lattice tower.

(13) The rotor blades **22** are spaced about the hub **20** to facilitate rotating the rotor **18** to enable kinetic energy to be transferred from the wind into usable mechanical energy, and subsequently, electrical energy. The rotor blades **22** are mated to the hub **20** by coupling a blade root region **24** to the hub **20** at a plurality of load transfer regions **26**. The load transfer regions **26** may have a hub load transfer region and a blade load transfer region (both not shown in FIG. 1). Loads induced to the rotor blades **22** are transferred to the hub **20** via the load transfer regions **26**.

(14) In examples, the rotor blades **22** may have a length ranging from about 15 meters (m) to about 90 m or more. Rotor blades **22** may have any suitable length that enables the wind turbine **10** to function as described herein. For example, non-limiting examples of blade lengths include 20 m or less, 37 m, 48.7 m, 50.2 m, 52.2 m or a length that is greater than 91 m. As wind strikes the rotor blades **22** from a wind direction **28**, the rotor **18** is rotated about a rotor axis **30**. As the rotor blades **22** are rotated and subjected to centrifugal forces, the rotor blades **22** are also subjected to various forces and moments. As such, the rotor blades **22** may deflect and/or rotate from a neutral, or non-deflected, position to a deflected position.

(15) Moreover, a pitch angle of the rotor blades **22**, i.e., an angle that determines an orientation of the rotor blades **22** with respect to the wind direction, may be changed by a pitch system **32** to control the load and power output by the wind turbine **10** by adjusting an angular position of at least one rotor blade **22** relative to wind vectors. Pitch axes **34** of rotor blades **22** are shown. During operation of the wind turbine **10**, the pitch system **32** may particularly change a pitch angle of the rotor blades **22** such that the angle of attack of (portions of) the rotor blades are reduced, which facilitates reducing a rotational speed and/or facilitates a stall of the rotor **18**.

(16) In the example, a blade pitch of each rotor blade **22** is controlled individually by a wind turbine controller **36** or by a pitch control system **80**. Alternatively, the blade pitch for all rotor blades **22** may be controlled simultaneously by said control systems.

(17) Further, in the example, as the wind direction **28** changes, a nacelle **16** may be rotated about the longitudinal axis of the tower, i.e. about a yaw axis **38** to position the rotor blades **22** with respect to wind direction **28**.

(18) In the example, the wind turbine controller **36** is shown as being centralized within the nacelle **16**, however, the wind turbine controller **36** may be a distributed system throughout the wind turbine **10**, on the support system **14**, within a wind farm, and/or at a remote-control center. The wind turbine controller **36** may include a processor **40** configured to perform some of the methods and/or steps described herein. Further, many of the other components described herein include a processor.

(19) As used herein, the term “processor” is not limited to integrated circuits referred to in the art as a computer, but broadly refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific, integrated circuit, and other programmable circuits, and these terms are used interchangeably herein. It should be understood that a processor and/or a control system can also include memory, input channels, and/or output channels.

(20) A control system **36** may also include a memory, e.g. one or more memory devices. A memory may comprise memory element(s) including, but not limited to, a computer readable medium (e.g., random access memory (RAM)), a computer readable non-volatile medium (e.g., a flash memory), a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD) and/or other suitable memory elements. Such memory device(s) may generally be configured to store suitable computer-readable instructions that, when implemented by the processor(s) **40**, configure the controller **36** to perform, or trigger the performance of, various steps disclosed herein. A memory may also be configured to store data, e.g. from measurements and/or calculations.

(21) FIG. **2** is an enlarged sectional view of a portion of the wind turbine **10**. In the example, the wind turbine **10** includes the nacelle **16** and the rotor **18** that is rotatably coupled to the nacelle **16**. More specifically, the hub **20** of the rotor **18** is rotatably coupled to an electric generator **42** positioned within the nacelle **16** by the main shaft **44**, a gearbox **46**, a high-speed shaft **48**, and a coupling **50**. In the example, the main shaft **44** is disposed at least partially coaxial to a longitudinal axis (not shown) of the nacelle **16**. A rotation of the main shaft **44** drives the gearbox **46** that subsequently drives the high-speed shaft **48** by translating the relatively slow rotational movement of the rotor **18** and of the main shaft **44** into a relatively fast rotational movement of the high-speed shaft **48**. The latter is connected to the generator **42** for generating electrical energy with the help of a coupling **50**. Furthermore, a transformer **90** and/or suitable electronics, switches, and/or inverters may be arranged in the nacelle **16** in order to transform electrical energy generated by the generator **42** having a voltage between 400V to 1000 V into electrical energy having medium voltage (e.g. 10-35 KV). Said electrical energy is conducted via power cables from the nacelle **16** into the tower **15**.

(22) The gearbox **46**, generator **42** and transformer **90** may be supported by a main support structure frame of the nacelle **16**, optionally embodied as a main frame **52**. The gearbox **46** may include a gearbox housing that is connected to the main frame **52** by one or more torque arms **55**. In the example, the nacelle **16** also includes a main forward support bearing **60** and a main aft support bearing **62**. Furthermore, the generator **42** can be mounted to the main frame **52** by decoupling support means **54**, in particular in order to prevent vibrations of the generator **42** to be introduced into the main frame **52** and thereby causing a noise emission source.

(23) Optionally, the main frame **52** is configured to carry the entire load caused by the weight of the rotor **18** and components of the nacelle **16** and by the wind and rotational loads, and furthermore, to introduce these loads into the tower **15** of the wind turbine **10**. The rotor shaft **44**, generator **42**, gearbox **46**, high speed shaft **48**, coupling **50**, and any associated fastening, support, and/or securing device including, but not limited to, support **52**, and forward support bearing **60** and aft support bearing **62**, are sometimes referred to as a drive train **64**.

(24) In some examples, the wind turbine may be a direct drive wind turbine without gearbox **46**. Generator **42** operate at the same rotational speed as the rotor **18** in direct drive wind turbines. They therefore generally have a much larger diameter than generators used in wind turbines having a gearbox **46** for providing a similar amount of power than a wind turbine with a gearbox.

(25) The nacelle **16** also may include a yaw system which comprises a yaw bearing (not visible in FIG. **2**) having two bearing components configured to rotate with respect to the other. The tower **15** is coupled to one of the bearing components and the bedplate or support frame **52** of the nacelle **16** is coupled to the other bearing component.

(26) The yaw system may comprise an annular gear **21** and a yaw drive mechanism **56** that may be

used to rotate the nacelle **16** and thereby also the rotor **18** about the longitudinal axis of the tower, i.e. about a yaw axis **38** to control the perspective of the rotor blades **22** with respect to the wind direction **28**.

(27) The yaw drive mechanism **56** may comprise a plurality of yaw drives **35** with a motor **33**, a gearbox **37** and a pinion **39** for meshing with the annular gear **21** for rotating one of the bearing components with respect to the other. The annular gear **21** may comprise a plurality of teeth which engage with the teeth of the pinion **39**. In the example of FIG. 2, the yaw drives **35** and the annular gear **21** are placed outside the external diameter of the tower. The teeth of the annular gear are outwardly orientated, but in other examples, the annular gear and yaw drives may be arranged at the inside of the tower.

(28) In some examples, one of the yaw drives may be a “master”, and the other drives may be “slaves” following the instructions of the master or adapting their operation to adapt to the master drive.

(29) The turbine controller **36** may be communicatively coupled to the yaw drive mechanism **56** of the wind turbine **10** for controlling and/or altering the yaw direction of the nacelle **16** relative to the wind direction **28**. As the direction of the wind **28** changes, the wind turbine controller **36** may be configured to control a yaw angle of the nacelle **16** about the longitudinal axis of the tower or yaw axis **38** to position the rotor blades **22**, and therefore the rotor **18**, with respect to the direction **28** of the wind, thereby controlling the loads acting on the wind turbine **10**. For example, the turbine controller **36** may be configured to transmit control signals or commands to the yaw drive mechanism **56** of the wind turbine **10**, via a yaw controller or direct transmission, such that the nacelle **16** may be rotated about the longitudinal axis of the tower or yaw axis **38** via a yaw bearing.

(30) For positioning the nacelle **16** appropriately with respect to the wind direction **28**, the nacelle **16** may also include at least one meteorological measurement system which may include a wind vane and anemometer. The meteorological measurement system **58** can provide information to the wind turbine controller **36** that may include wind direction **28** and/or wind speed.

(31) In the example, the pitch system **32** is at least partially arranged as a pitch assembly **66** in the hub **20**. The pitch assembly **66** includes one or more pitch drive systems **68** and at least one sensor **70**. Each pitch drive system **68** is coupled to a respective rotor blade **22** (shown in FIG. 1) for modulating the pitch angle of a rotor blade **22** along the pitch axis **34**. Only one of three pitch drive systems **68** is shown in FIG. 2.

(32) In the example, the pitch assembly **66** includes at least one pitch bearing **72** coupled to hub **20** and to a respective rotor blade **22** (shown in FIG. 1) for rotating the respective rotor blade **22** about the pitch axis **34**. The pitch drive system **68** includes a pitch drive motor **74**, a pitch drive gearbox **76**, and a pitch drive pinion **78**. The pitch drive motor **74** is coupled to the pitch drive gearbox **76** such that the pitch drive motor **74** imparts mechanical force to the pitch drive gearbox **76**. The pitch drive gearbox **76** is coupled to the pitch drive pinion **78** such that the pitch drive pinion **78** is rotated by the pitch drive gearbox **76**. The pitch bearing **72** is coupled to pitch drive pinion **78** such that the rotation of the pitch drive pinion **78** causes a rotation of the pitch bearing **72**.

(33) Pitch drive system **68** is coupled to the wind turbine controller **36** for adjusting the pitch angle of a rotor blade **22** upon receipt of one or more signals from the wind turbine controller **36**. In the example, the pitch drive motor **74** is any suitable motor driven by electrical power and/or a hydraulic system that enables pitch assembly **66** to function as described herein. Alternatively, the pitch assembly **66** may include any suitable structure, configuration, arrangement, and/or components such as, but not limited to, hydraulic cylinders, springs, and/or servomechanisms. In certain embodiments, the pitch drive motor **74** is driven by energy extracted from a rotational inertia of hub **20** and/or a stored energy source (not shown) that supplies energy to components of the wind turbine **10**.

(34) The pitch assembly **66** may also include one or more pitch control systems **80** for controlling the pitch drive system **68** according to control signals from the wind turbine controller **36**, in case

of specific prioritized situations and/or during rotor **18** overspeed. In the example, the pitch assembly **66** includes at least one pitch control system **80** communicatively coupled to a respective pitch drive system **68** for controlling pitch drive system **68** independently from the wind turbine controller **36**. In the example, the pitch control system **80** is coupled to the pitch drive system **68** and to a sensor **70**. During normal operation of the wind turbine **10**, the wind turbine controller **36** may control the pitch drive system **68** to adjust a pitch angle of rotor blades **22**.

(35) According to an embodiment, a power generator **84**, for example comprising a battery and electric capacitors, is arranged at or within the hub **20** and is coupled to the sensor **70**, the pitch control system **80**, and to the pitch drive system **68** to provide a source of power to these components. In the example, the power generator **84** provides a continuing source of power to the pitch assembly **66** during operation of the wind turbine **10**. In an alternative embodiment, power generator **84** provides power to the pitch assembly **66** only during an electrical power loss event of the wind turbine **10**. The electrical power loss event may include power grid loss or dip, malfunctioning of an electrical system of the wind turbine **10**, and/or failure of the wind turbine controller **36**. During the electrical power loss event, the power generator **84** operates to provide electrical power to the pitch assembly **66** such that pitch assembly **66** can operate during the electrical power loss event.

(36) In the example, the pitch drive system **68**, the sensor **70**, the pitch control system **80**, cables, and the power generator **84** are each positioned in a cavity **86** defined by an inner surface **88** of hub **20**. In an alternative embodiment, said components are positioned with respect to an outer surface of hub **20** and may be coupled, directly or indirectly, to the outer surface.

(37) FIG. **3** illustrates a portion of yaw system according to one example. The yaw system comprises a yaw bearing **98** having a first bearing component **95** coupled to the tower **15** and a second bearing component **96** coupled to the main frame **52** of the nacelle. The yaw system comprises an annular gear **21** which, in this example, is coupled to the first bearing component **95**. Other configurations are also possible such as, e.g. the annular gear **21** may be coupled to the tower **15**. The yaw system comprises a yaw drive **35** and the yaw drive **35** in this example comprises a motor **33**, a gearbox **37** and a pinion **39** for meshing with the annular gear **21**. The rotation of the motor may rotate the pinion with respect to the annular gear **21**. As mentioned, in this example the annular gear **21** is coupled to the first bearing component **95**, so the first bearing component **95** may rotate with respect to the second bearing component **96** which is coupled to the main frame **52**. Although only a single yaw drive is shown, it should be clear that multiple yaw drives, each with their motor, gearbox and pinion may be provided around the circumference of the tower.

(38) The annular gear **21** may comprise a plurality of teeth which engage with the teeth of the pinion **39**. In this example, the yaw drive **35** and the annular gear **21** are placed outside the external diameter of the tower. The teeth of the annular gear **21** are outwardly oriented. However, in other examples, the annular gear **21** and the yaw drive **35** may be placed inside the tower.

(39) The yaw system of FIG. **3** further comprises a braking disk **93** and a hydraulic brake unit **94** configured to exert a braking force on the braking disk **93** for braking the rotation of the first bearing component **95** with respect to the second bearing component **96**. Although only a single hydraulic brake unit is shown, a plurality of brake units may be provided along the circumference of the tower.

(40) In FIG. **3**, the braking disk **93** is connected to the tower **15**. In other examples, the braking disk **93** may be integrally formed with the tower **15**. The hydraulic brake unit **94** may comprise a brake caliper attached to the support frame **52** and may engage the brake disk **93** to prevent rotation of the nacelle **16** with respect to the tower **15**. In order to activate the brake unit **94**, i.e. to engage with the brake disk **93**, the hydraulic brake unit **94** may comprise an active hydraulic system to control the pressure of the brake caliper on the braking disk **93**.

(41) In some examples, the actuation of the brake unit **94** may be controlled by a controller **92**. In some examples, the controller **92** may be embodied in the wind turbine controller. The controller

92 may activate or deactivate the brake unit **94** or it may actuate the brake unit **94** to control the pressure of the hydraulic system so as to control the braking moment exerted by the brake unit **94** when engaging with the brake disk **93**. In particular, a moderate braking moment (e.g. in the range of 20-40% depending on the design and mode of operation) may be exerted during the rotation of the nacelle to provide some damping **16** and an actuation may be needed to increase the braking moment exerted by the brake unit **94** to stop the yaw operation.

(42) In some examples, the motor **33** may comprise a motor brake **91**. The motor brake **91** may be arranged inside the housing of the motor **33** and it may be arranged to block rotation of the motor **33**. The motor brake **91** may comprise a friction brake where a friction disk is secured to the rotatable shaft of the electric motor **33**. The friction disk may be configured to engage a non-rotating surface, thus exerting a braking moment on the system.

(43) A control system **92** configured to control the yaw system according to a certain algorithm is also illustrated in FIG. 3. In this example, a stand-alone control system **92** is depicted.

Nevertheless, it is understood that the control of the yaw system may be implemented in any controller, including, e.g. the wind turbine controller **36**.

(44) The yaw system of FIG. 3 comprises a bearing **98** of the roller bearing type. Nevertheless, other types of bearings such as gliding yaw bearings are well-known in the art and may be also used in further examples.

(45) The motor illustrated in FIG. 3 may be directly connected to the power supply or it may be connected to the power supply via a vertical frequency drive, VFD (not shown in the figure). The use of such VFDs is well-known in the art so no further details are herein provided.

(46) FIG. 10 depicts the time evolution of the yaw operation according to a method of the prior-art. In particular, the behavior of the system at an end of the yaw operation is depicted. Line **901** shows the Yaw Control command, which transitions from a high level to a low level at t.sub.END to indicate an end condition for the yaw operation. Line **902** shows the status of the yaw motor **33** which is stopped at the end of the yaw operation so that no more driving moment is applied to rotate the nacelle with respect to the tower. The hydraulic brake **94**, whose evolution is represented by line **904**, is actuated at t.sub.END to help stop the rotation by applying a corresponding braking moment. Finally, the motor brake **91** is also activated after the end of the yaw operation after some inevitable delay. The actuation of the hydraulic brake **94** represented in line **904** may not necessarily involve an “activation” (i.e. turning the brake “on”) of the hydraulic brake **94**. In some applications, the hydraulic brake **94** may already exert a moderate braking moment during the yaw operation so that the transition in line **904** would represent a change of the operating state (but not necessarily an activation) of the hydraulic brake **94** to increase its braking moment.

(47) As mentioned in the introduction, it has been found that this prior-art method can lead to misalignment and premature wear of the yaw system. It has been observed that the misalignment results in an increased number of yaw operations for the wind turbine and, in some cases, it requires shut down of the wind turbine, which increases economic revenue and further increases the wear of the components.

(48) After careful analysis, the inventors identified a problem at the end of the yaw operation. Yaw motors **33** are deactivated at t.sub.END so control of the nacelle **16** by the motors **33** is no longer possible. On the other hand, even if the hydraulic brakes **94** are activated at t.sub.END, it has been found that a period exists during which the braking capacity of the hydraulic brakes **94** is insufficient to control the movement of the nacelle **16**. During that timeframe, the nacelle **16** is still subject to certain loads, such as wind loads, and it may slip, especially in case of high wind loads due to. e.g. high prevailing wind speeds or the occurrence of a wind gust. Inventors have found that such slipping may lead to uncontrolled nacelle acceleration.

(49) As a result of this movement of the nacelle **16**, the yaw speed (line **905**) may suffer uncontrolled deviations as represented by the dashed line **905** after t.sub.END. It has been found that these nacelle accelerations may result in further movement of the nacelle after t.sub.END

leading to significant yaw misalignment or to significant wear or damage of wind turbine components such as components of the yaw system itself or even other components of the wind turbine in case of very high slip resulting in mechanical loads exceeding design loads. As an example, the motor brake **91** may be actuated at t.sub.MOT but, depending on prevailing conditions, the yaw speed at t.sub.MOT may be excessive, which may induce premature wear of the motor brake **91**.

(50) Based on these observations, the inventors have developed new control methods for yaw operation as explained below.

(51) In an example of the disclosure, a method **100** for controlling a wind turbine **10** during a yaw operation is provided. The method **100** is schematically illustrated in the flow chart of FIG. **4**. The method comprises, at block **110**, operating the yaw drives **35** to rotate the nacelle **16** with respect to the tower **15**. The method further comprises, at block **120**, predicting an end of the yaw operation. Block **130** of the method comprises, in response to predicting the end of the yaw operation in block **120**, actuating the hydraulic brake **94** before the predicted end.

(52) It should be noted that, actuating the hydraulic brake may either result in the activation of the hydraulic brake or in a modification on the operating conditions of the hydraulic brake. As mentioned before, it is known from examples of the prior art, that a certain braking moment can be applied by a hydraulic brake during the yawing operation to provide some damping. Thus, it is known to operate the hydraulic brakes at 20-40% of its rated moment while yawing during power production, so that damping is achieved in some existing designs. In such cases, the actuation of the hydraulic brake according to the present disclosure, which is carried out in response to predicting an end of the yaw operation itself, would result in a change, and more specifically in an increase, of the braking moment above the braking moment exerted during the yaw rotation.

(53) In some other modes of operation, i.e. when yawing for cable untwisting, lower values (or even 0%) may be used during the yawing. In those cases, actuation according to this aspect may result in the activation of the hydraulic brake.

(54) Specific values depend on certain design details of the yaw system in the wind turbine. For example, yaw systems comprising gliding bearings may achieve an adequate level of damping during power production mode with less pressure (or even with no pressure) on the hydraulic brakes than yaw systems comprising ball bearings.

(55) This example of the disclosure will now be explained with reference to FIG. **5**, which illustrates the time evolution of the yaw operation according to the method **100** as well as the evolution of the nacelle rotation speed during the yaw operation. Line **101** shows the Yaw Control command that is used to initiate the yaw operation (by switching from low to high in the figure) or to indicate that an end condition for the yaw operation has been reached, in which case the Yaw Control signal is switched from high to low. This command is provided by a controller **92** that controls yaw operation of the wind turbine and which can be either a stand-alone controller or a controller implemented in the wind turbine controller **36**.

(56) Line **102** shows the status of the yaw motor **33**. The yaw motor **33** is activated when the Yaw Control signal is switched from low to high so that a driving moment is applied by the yaw drive **35** to actively rotate the nacelle **16** with respect to the tower **15** and initiate the yaw operation. Line **103** shows the behavior of the motor brake **91** which is also deactivated when the yaw system receives the indication to initiate the yaw operation to allow the yaw operation to take place. Also illustrated in this example is the evolution **104** of the hydraulic brake **94**, the pressure of which is reduced when starting the yaw operation so that the braking moment exerted by the hydraulic brake **94** is reduced. Finally, **105** schematically represents the yaw speed, i.e. the rotational speed of the nacelle **16** as it rotates around the longitudinal axis of the tower **15**. As expected, activation of the Yaw Control signal **101** results in an increase of the yaw speed **104** for a brief period until a stabilized speed is reached for the yaw operation.

(57) According to this example of the disclosure, an end of the yaw operation is predicted during

the yaw operation. A high frequency can be used for the prediction, which in specific examples comprises a cycle time for the prediction between 20-100 ms and, more specifically, of 40 ms during the yaw operation.

(58) In the example shown in FIG. 5, the end of the yaw operation occurs at a time $t_{sub.END}$. According to this example of the method and, more particularly, according to block **130** of the method **100**, the hydraulic brake **94** is actuated at an earlier time, $t_{sub.HYD}$, as seen in curve **104** so that the braking moment is increased at an early stage and it reaches a significant braking moment by the time $t_{sub.END}$. The actuation of the hydraulic brake **94** occurring at $T_{sub.HYD}$ is made in response to the prediction of the end of the yaw operation. According to this example, the braking moment resulting from the early activation of the hydraulic brake **94** reduces the potential slip of the nacelle **16** occurring during the finalization of the yaw operation.

(59) In an example of a method for operating a wind turbine, predicting an end of the yaw operation comprises predicting a time for finalization of the yaw operation and/or predicting an orientation of the nacelle **16** with respect to the tower **15** for finalization of the yaw operation. In particular, as further detailed below in relation to some specific examples, different methods may be used for predicting the end of the yaw operation which may depend on, e.g. the reason for the yaw operation. In some circumstances, the end of the yaw operation may be better defined by a specific time (or time left with respect to the current time) by which the yaw operation is expected to end or by the specific orientation that the nacelle **16** is expected to exhibit at the end of the yaw operation.

(60) In an example of a method, the hydraulic brake **94** is characterized by a rated braking moment. The method comprises actuating the hydraulic brake **94** in a manner and at a time, $t_{sub.HYD}$, such that the braking moment exerted at $t_{sub.END}$ is at least 60%, specifically 70% or more specifically at least 80% of the rated value. With increasing values of the braking moment, the potential slipping of the nacelle is further reduced and, consequently, the potential risk of damage or failure of wind turbine components is further mitigated.

(61) In another example of a method for operating a wind turbine during a yaw operation, the evolution of the pressure of the hydraulic brake **94** is considered. Thus, a hydraulic brake **94** may be defined by the time needed for its pressure to reach a predetermined value, starting from an operational situation i.e. the hydraulic brake pressure rise time. In particular, the predetermined value may refer to a percentage of the hydraulic pressure rated value such as, e.g. 80%, 90% or 100% of the rated value. Thus, in this example, actuating the hydraulic brake **94** at $t_{sub.HYD}$ before the predicted end time, i.e. before $t_{sub.END}$, may comprise actuating the hydraulic brake **94** based on the hydraulic brake pressure rise time. In a more particular example, actuating the hydraulic brake **94** based on the hydraulic brake pressure rise time comprises actuating the brake **94** when the time left until the predicted end is substantially equal to the hydraulic brake pressure rise time. In other words, upon predicting the time $t_{sub.END}$, $t_{sub.HYD}$ is selected so that $t_{sub.END} - t_{sub.HYD}$ is substantially equal to the hydraulic brake pressure rise time. With this example, the braking moment applied at $t_{sub.END}$ by the hydraulic brake **94** is close to the value expected for the predetermined value of the hydraulic pressure, which may be, for instance, 80%, 90 or 100% of its rated value. This would result in a high braking moment being applied so that nacelle **16** slipping and associated risks are highly mitigated.

(62) The hydraulic brake pressure rise time may be determined as the time needed for the pressure to reach a predetermined value (e.g. a percentage of the rated value) when starting from a value of the pressure corresponding to the operation of the hydraulic brake **94** during yaw operation. In some examples, the hydraulic brake **94** may be deactivated during rotation. But in some other cases, the hydraulic brake **94** may be exerting a moderate braking moment, e.g. 40% of its rated value, for damping purposes. In the former case, the hydraulic pressure rise time would correspond to the time needed to increase the pressure from 0% of the predetermined value (e.g. a percentage of the rated value) whereas in the latter case, the hydraulic pressure rise time would correspond to

the time needed to increase the pressure from 40% of the rated value to the predetermined value.

(63) In an example of a method for operating a wind turbine, predicting an end of the yaw operation comprises predicting the time when the nacelle **16** will reach a predetermined orientation with respect to the tower **15**. This may be the case for some yaw operations involving deterministic yaw movements such as, e.g. a movement related to a cable untwisting operation in which the final orientation of the nacelle **16** with respect to the tower **15** is predetermined based on the arrangement of the cables with respect to the tower **15**. In such cases, the orientation of the nacelle **16** at the end of the yaw operation may be known but the time needed to reach that position may need to be predicted as this may be dependent on different design and/or prevailing operating conditions.

(64) In yet a further example, predicting an end of the yaw operation comprises predicting the time and/or the nacelle **16** orientation with respect to the tower **15** at which an error between the orientation of the nacelle **16** and a prevailing wind direction **28** is equal to or below a threshold value. This may be the case in certain modes of operation, such as power production mode, involving non-deterministic movements of the nacelle **16**. Indeed, it is known that, especially during power production mode, the objective is for the rotor **18** of the wind turbine **10** to be aligned with the prevailing wind direction. Nevertheless, wind direction **28** is constantly changing so a varying yaw error is defined in real time as the difference between the wind direction **28** and the actual orientation of the nacelle **16**.

(65) The wind direction **28** can be measured with a wind vane, estimated from load measurements, or obtained in other manners known in the art. As the wind direction **28** can change rapidly, average values, e.g. obtained over ten-minute periods, may be used. In order to ensure optimum energy capture from the wind and proper loads control, a yaw error threshold may be defined which may not be exceeded. Particularly, a yaw error threshold may be defined for activation of the yaw system. In case the determined yaw error exceeds said threshold, activation of the yaw system may be carried out in order to reduce the misalignment. Similarly, a yaw error threshold may be used to indicate the end of a yaw operation after orientation of the nacelle **16** reduces the misalignment of the nacelle **16** with respect to the wind direction **28** to a sufficiently low value. In some examples, hysteresis may be provided so that the value of the yaw error threshold for activation of the yaw system is larger than the value of the yaw error threshold for indication of an end condition.

(66) In some examples, the actual wind direction **28**, the actual orientation of the nacelle **16** with respect to the tower **15** and the speed of rotation of the nacelle **16** may be used to predict the end time or the nacelle orientation relative to the tower **15** at the end of the yaw operation. In particular, the actual wind direction **28** and the actual orientation of the nacelle **16** may be used to derive the actual yaw error. The evolution over time of the yaw error may then be used to predict the time and/or the corresponding orientation of the nacelle **16** by the time the yaw error satisfies the threshold condition to end the yaw operation.

(67) In another example of a method to operate a wind turbine during a yaw operation, the yaw drive **35** comprises at least one motor **33** which is turned off at t.sub.END, i.e. when the end of the operation is reached. By stopping the motors **33**, no more driving moment is applied to the nacelle **16** so that the movement of the nacelle **16** is determined by the braking moment applied by the braking system and by the loads acting on the wind turbine **10**. In some examples, the at least one motor **33** may be turned off after a certain delay after t.sub.END. This delay may be useful to provide some additional time for the hydraulic brake **94** to build its pressure and braking moment. Nevertheless, the delay may be kept short to prevent any noticeable yaw offset with respect to the initially intended orientation.

(68) In yet further examples, the yaw drive **35** comprises one or more yaw motors **33** that comprise a motor brake **91**. In those examples, the motor brake **91** may be activated after the end time of the yaw operation, t.sub.END, and after stopping the yaw motors **33**. Such behavior is also presented

in FIG. 5 which depicts the activation of the motor brake at $t_{sub.MOT}$. Activation of the motor brake **91** may be triggered differently according to different examples. Motor brakes may comprise some friction elements which may wear prematurely if activation of the motor brake **91** occurs while rotational speed is high. For that reason, it may be preferred to activate the motor brake **91** once the rotational speed has decreased significantly. In that sense, an example of the invention may comprise activating the motor brake **91** after a predetermined delay after the end of the yaw operation. In that case, $t_{sub.MOT}$ and $t_{sub.END}$ may be separated by some predetermined time delay which may be selected under the estimation that, at least under some conditions, the rotational speed by the time $t_{sub.MOT}$ is sufficiently low or even null.

(69) FIG. 6 shows a flow chart of an example of another method **200** for controlling a wind turbine **10** during a yaw operation and, more particularly, for controlling a yaw motor **33** and a motor brake **91** during an end of a yaw operation. In this method, the activation of the yaw motor brake **91** is not based on a certain predetermined time but on the nacelle **16** speed. The method comprises, at block **210**, receiving an indication of an end of the yaw rotation operation. This indication may be generated by a controller **92** and this may be based on different criteria depending on the mode of operation as explained above. Block **220** comprises stopping the yaw motors **33** so that no more driving moment is provided in an active manner. Block **230** comprises determining the nacelle speed whereas block **240** comprises checking if the nacelle speed reaches a predetermined minimum speed threshold ($v_{sub.MIN}$). The method remains at block **240** while the nacelle **16** speed is above $v_{sub.MIN}$. Block **250** corresponds to the situation when the nacelle **16** speed reaches the minimum speed threshold, in which case the motor brake **91** is activated.

(70) In order to further illustrate this method, FIG. 7 schematically shows the evolution of the Yaw Control command **201**, the yaw motor status **202**, the yaw motor brake status **203**, the yaw speed of the nacelle **204** during the operation and the variation of the yaw orientation **205** of the nacelle **16** with respect to the tower **15** during the operation. As already explained in regard to FIG. 5, the yaw speed **204** is increased upon activation of the yaw operation until a stable yaw speed is reached. The yaw operation ends at $t_{sub.END}$ as indicated by the Yaw Control command **201** switching from high to low. According to this example, the yaw motor **33** is stopped as also indicated by the corresponding transition from high to low in curve **202**. The nacelle speed is then determined. Two different scenarios, resulting in different evolutions of the nacelle speed, are presented in FIG. 7. Thus, both curves **204a** and **204b** exhibit the expected behavior during the braking process, i.e. the nacelle **16** speed is reduced, but at different rates.

(71) As previously mentioned, it is preferable to activate the motor brake **91** at a low speed to prevent premature wear of friction parts. For that reason, a minimum speed threshold, $v_{sub.MIN}$, is predetermined which the aim to maintain the wear of the motor brake within acceptable design parameters. In this particular example, curve **204a** presents a situation where the nacelle speed decreases rapidly after $t_{sub.END}$. Consequently, the activation time for the motor brake **91** in this situation ($t_{sub.MOT_a}$) is very close to the end time for the yaw operation, $t_{sub.END}$.

(72) On the other hand, curve **204b** presents a scenario where the slowing down of the nacelle takes more time so that the nacelle speed remains above the minimum threshold for a longer time. In this second case, the motor brake **91** is not activated until $t_{sub.MOT_b}$. Note that this has an effect also on the final orientation of the nacelle. Thus, curves **205a** and **205b** show the evolution of the yaw position after the end of the yaw operation. It can be observed how the yaw deviation is lower (**205a**) in the case where the nacelle speed reaches the minimum threshold in a short period of time whereas a larger deviation (**205b**) on the yaw position can be observed in the scenario where the nacelle **16** speed remains above the minimum threshold for activation of the motor brake **91** for a longer time. The change from one scenario to the other may be dependent on different factors, including but not limited to, the prevailing wind conditions during the stop sequence or the braking moment exerted by other braking mechanisms that may be present in the wind turbine **10**.

(73) FIG. 8 shows a flow chart of still a further example of a method **300** for controlling a wind

turbine **10** during a yaw operation and, more particularly, for operating a yaw motor **33** and a motor brake **91** during an end of a yaw operation. A block **310** is included which comprises, similarly, to block **210** of method **200**, receiving an indication of an end of the yaw rotation operation. Block **320** comprises stopping the yaw motors **33**. Block **330** comprises determining the yaw speed of the nacelle and/or the yaw acceleration of the nacelle whereas block **340** comprises comparing those determined variables with some predetermined maximum thresholds. If the determined yaw speed or the determined yaw acceleration exceeds the respective predetermined threshold, the method continues to block **350**, which comprises activating the motor brake **91** so that further movement of the nacelle **16** is limited.

(74) In this further example of a method for operating a wind turbine, which is also schematically illustrated by the time evolution of the different elements in FIG. **9**, a different behavior than in the previous example is considered. In particular, curves labelled with an “a” or a “b” in FIG. **9** depict different scenarios where the nacelle **16** is not slowing down at the end of the yaw operation as typically expected. On the contrary, the nacelle **16** in these examples experiences increased speed (**304a**) or acceleration (**305b**) after the end of the yaw operation and the stopping of the yaw motor **33**. Thus, solid lines (corresponding to a first “a” scenario) show a situation where the nacelle speed **304a** reaches a certain predetermined maximum threshold, $v_{sub.MAX}$, whereas dashed lines (corresponding to a second “b” scenario) show a situation where a sudden increase of the acceleration is observed at the end of the yaw operation, which results in the nacelle acceleration (**305b**) reaching a predetermined maximum threshold, $a_{sub.MAX}$.

(75) Different physical situations may lead to such behavior, including but not limited to a very high wind gust occurring at that moment or the malfunctioning of other braking mechanisms such as, e.g. a hydraulic brake **94**, in the wind turbine **10**. In such situations, this further example shows activation of the motor brake **91** even if the rotational speed is high. As mentioned before, this is not a preferred mode of operation for the motor brake **91** as operation at high speeds may induce significant wear or damage on the friction system of the motor brake **91**. Nevertheless, this example is dealing with certain scenarios where the motor brake **91** may be used as a safety system for the wind turbine **10**.

(76) FIG. **9** also depicts the evolution of the yaw position **306** in the two above mentioned scenarios. This schematically illustrates how the orientation of the nacelle **16** with respect to the tower **15** exhibits some runaway resulting from the extreme speed and/or acceleration. Nevertheless, the magnitude of the runaway is contained by the activation of the motor brake **91**.

(77) Even if FIG. **9** refers to an example of a method for operating a wind turbine **10** in which both the nacelle speed and the nacelle acceleration may be considered, further examples are also included which may comprise using only the speed or only the acceleration to detect operating conditions for which the motor brake **91** may be activated as a safety mechanism.

(78) Nacelle speed and nacelle acceleration may refer to different directions of rotation of the nacelle **16** with respect to the tower **15**. In particular, the direction of said rotation may be dependent on the loads acting on the wind turbine **10** after the predicted time to end the yaw operation, $t_{sub.END}$. In variants of the above-mentioned methods **200** and **300**, the respective thresholds for nacelle speed and nacelle acceleration for activation of the motor brake **91** may be dependent on the direction of the rotation of the nacelle **16** and, more particularly, on whether the rotation experienced after $t_{sub.END}$ is in the same direction as the rotation during the yaw operation.

(79) Note that the evolution of the different magnitudes presented in FIGS. **5**, **7** and **9** and, more particularly, the evolution of the nacelle speed, nacelle acceleration and nacelle orientation are provided for illustrative purposes only to help understand the methods but they should not be understood as accurately representing the operation of the nacelle.

(80) Details and explanations of methods **100**, **200** and **300** may be combined with and applied to each other.

(81) According to another aspect of the disclosure, a controller **92** is provided. The controller **92** is configured to perform the methods of operating a wind turbine during a yaw operation as described throughout this disclosure or any equivalent examples (including combinations). The controller **92** may be implemented as a stand-alone controller to control the yaw system or it may be implemented as part of the wind turbine controller **36**.

(82) The controller **92** may be configured to receive a plurality of signals **501**, such as signals indicative of wind direction **28**, nacelle **16** orientation with respect to the tower **15**, nacelle speed, nacelle acceleration, etc. The controller **92** may use that information to obtain a plurality of signals **502** to control the yaw system including generating the Yaw Control command (**101**, **201**, **301**) and controlling the hydraulic brake **94**, the yaw motors **33** or the motor brakes **91**.

(83) According to another aspect of the disclosure, a wind turbine **10** comprising a tower **15**, a nacelle **16** rotatably mounted on top of the tower **15**, a wind turbine rotor **18** including a plurality of wind turbine blades **22** and a yaw system is provided. The wind turbine comprises a controller **92** that is configured to perform the methods as described throughout this disclosure or any equivalent examples (including combinations). In some examples, the wind turbine **10** may comprise a yaw system with variable frequency drives (VFDs) to control the yaw motors **33**. The use of such VFDs may provide additional control capabilities to the yaw operation.

(84) This written description uses examples to disclose the teaching, and also to enable any person skilled in the art to practice the teaching, including making and using any devices or systems and performing any incorporated methods. The patentable scope is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims. Aspects from the various examples described, as well as other known equivalents for each such aspects, can be mixed and matched by one of ordinary skill in the art to construct additional examples and techniques in accordance with principles of this application. If reference signs related to drawings are placed in parentheses in a claim, they are solely for attempting to increase the intelligibility of the claim, and shall not be construed as limiting the scope of the claim.

Claims

1. A method for controlling a wind turbine during a yaw operation, the wind turbine including a nacelle, a tower, and a yaw system that rotates the nacelle relative to the tower about to a longitudinal axis of the tower, the yaw system including a hydraulic brake and one or more yaw drives, the method comprising: operating the one or more yaw drives at a first operational state in a yaw operation to rotate the nacelle with respect to the tower; predicting an end of the yaw operation when the one or more yaw drives are stopped; in response to the predicted end of the yaw operation, actuating the hydraulic brake before the predicted end of the yaw operation so that a predetermined braking moment of the hydraulic brake is achieved at or before the predicted end of the yaw operation; and maintaining the one or more yaw drives at the first operational state until the predicted end of the yaw operation.
2. The method according to claim 1, wherein the predicting an end of the yaw operation comprises predicting a time of completion of the yaw operation or a nacelle orientation with respect to the tower for completion of the yaw operation.
3. The method according to claim 1, wherein the hydraulic brake has a rated braking moment, and wherein the predetermined braking moment exerted by the hydraulic brake at the predicted end of the yaw operation is at least 60% of the rated braking moment.
4. The method according to claim 1, wherein, after actuation, the hydraulic brake has a rise time before reaching the predetermined braking moment, wherein the actuating the hydraulic brake

before the predicted end of the yaw operation comprises actuating the hydraulic brake based on the rise time.

5. The method according to claim 4, wherein the hydraulic brake is actuated so that the hydraulic brake is at the predetermined braking moment at the predicted end of the yaw operation.

6. The method according to claim 1, wherein the predicting an end of the yaw operation comprises predicting a time for an orientation of the nacelle with respect to the tower.

7. The according to claim 1, wherein the predicting an end of the yaw operation comprises predicting a time of completion of the yaw operation or time of a nacelle orientation with respect to the tower for completion of the yaw operation at which an error between the orientation of the nacelle and a prevailing wind direction is equal to or below a threshold value.

8. The method according to claim 7, wherein the error between the orientation of the nacelle and a prevailing wind direction is based on an actual wind direction, an actual orientation of the nacelle with respect to the tower, and the speed of rotation of the nacelle.

9. The method according to claim 1, wherein the one or more yaw drives comprise a yaw motor and a motor brake, the method comprising stopping the yaw motor at the predicted end of the yaw operation.

10. The method according to claim 9, the method comprising activating the motor brake after stopping the yaw motor.

11. The method according to claim 10, comprising activating the motor brake after a predetermined delay after the predicted end of the yaw operation or after the yaw motor is stopped.

12. The method according to claim 10, further comprising determining a speed of the nacelle and activating the motor brake when the nacelle speed reaches a predetermined minimum speed threshold.

13. The method according to claim 10, further comprising determining an acceleration of the nacelle and activating the motor brake when the nacelle acceleration reaches a predetermined maximum acceleration threshold.

14. A control system for a wind turbine, wherein the control system is configured to perform the method according to claim 1.

15. A wind turbine, comprising a control system configured to perform the method according to claim 1.
